

Human Performance Modeling - 3

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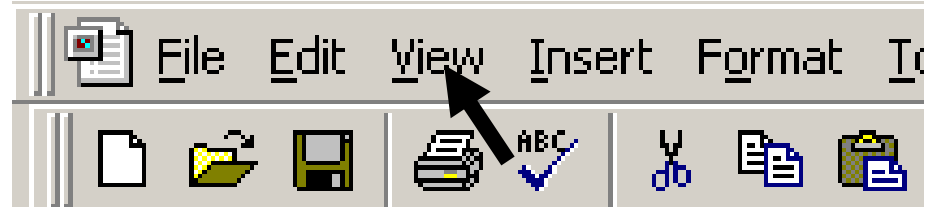
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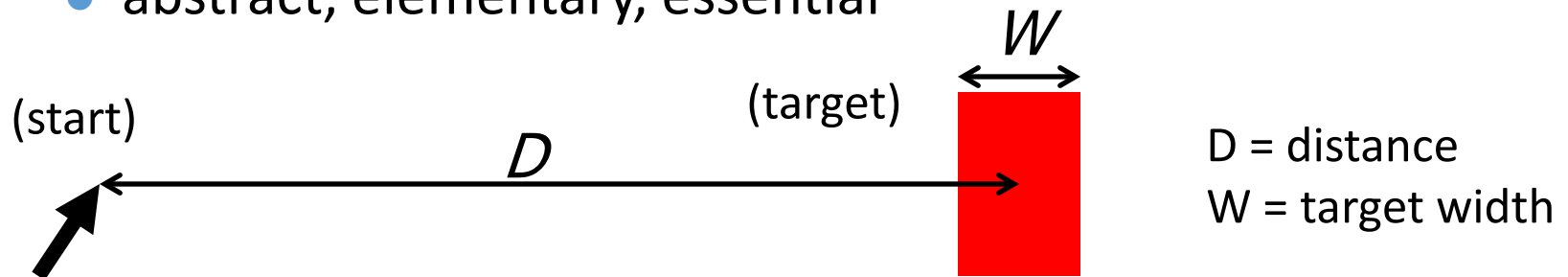
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Pointing Tasks

- Real task: interacting with GUI's
 - Pointing is fundamental



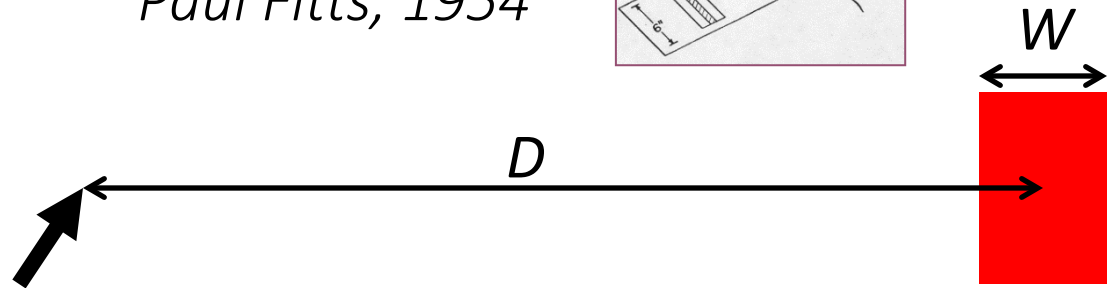
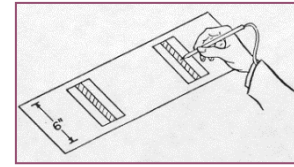
- Experimental task: target acquisition
 - abstract, elementary, essential



D is also referred to as A (amplitude) in literature.

Fitts' Law

Paul Fitts, 1954



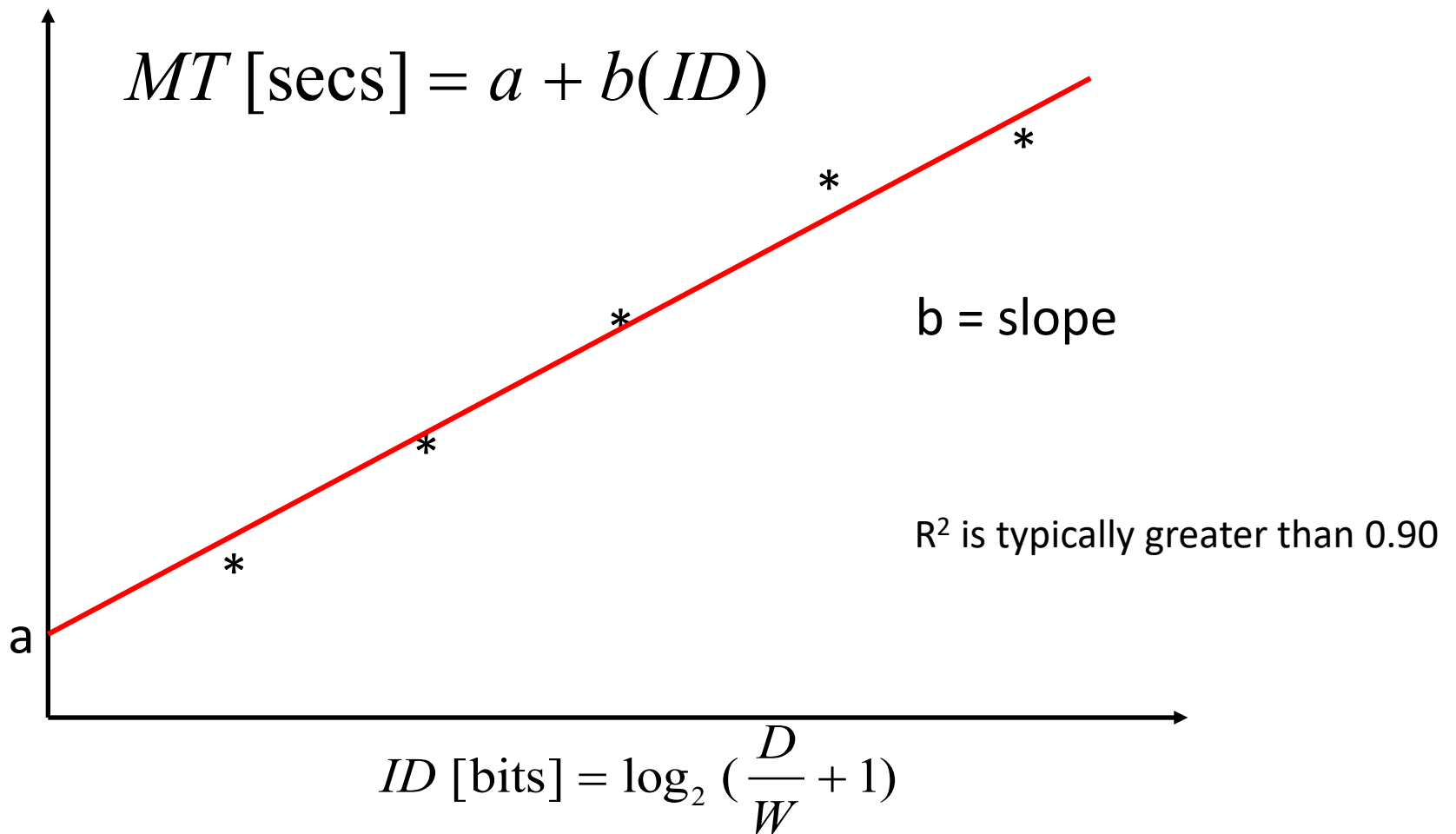
$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Movement Time

Index of Difficulty ($ID [bits]$)

The original Fitts' law proposed by Paul Fitts in 1954:

$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Only Relative Precision (D / W) Matters

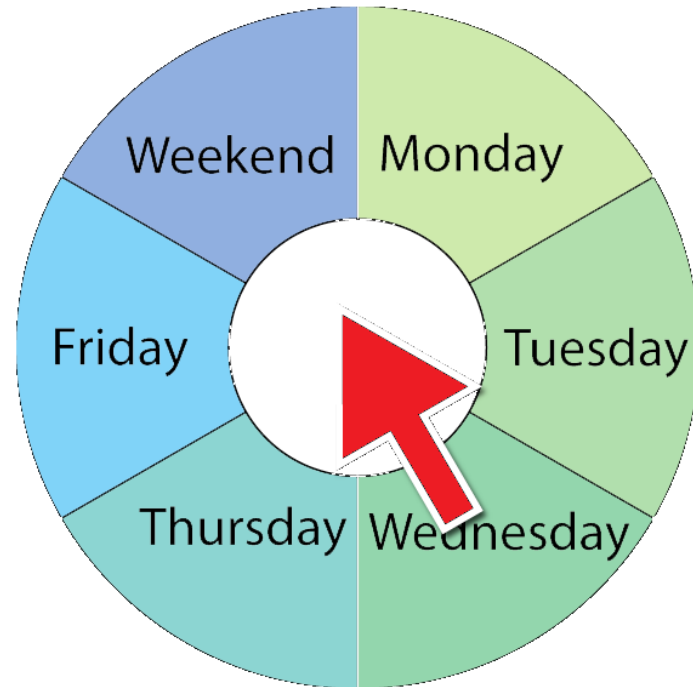
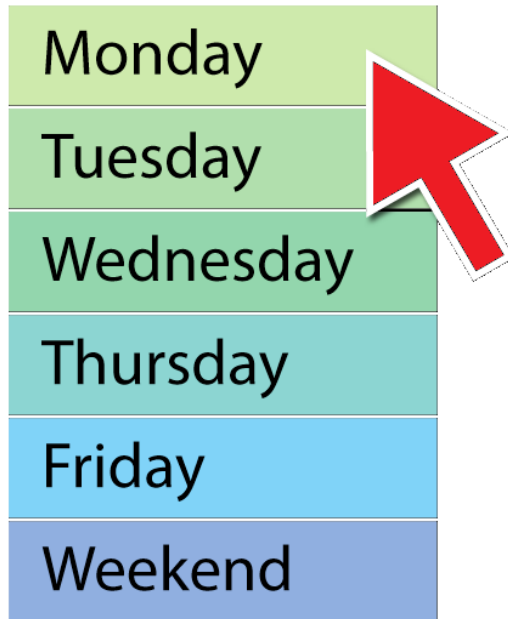
Explanation of R^2

$$R^2 \equiv 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2$$

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2,$$

Fitts' Law Example



Which will be faster on average?

pie menu (bigger targets & less distance)

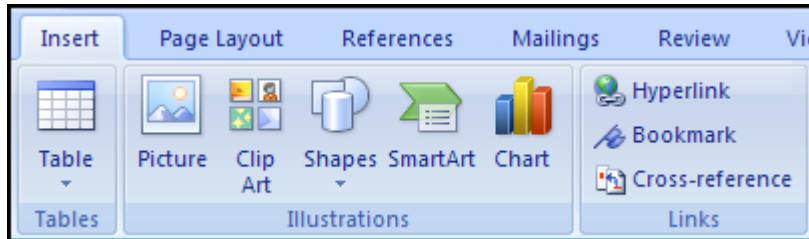
Fitts' Law in Windows and Mac OS



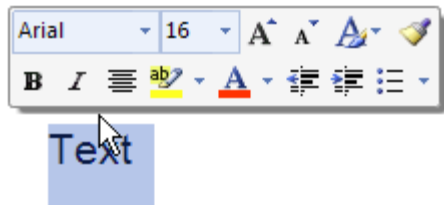
Windows 95: Missed by a pixel



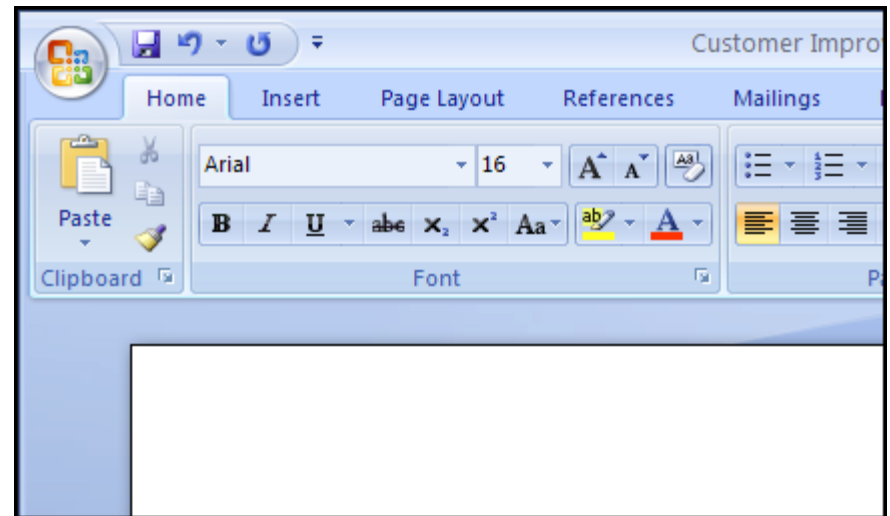
Fitts' Law in Office 2007



Larger, labeled controls can be clicked more quickly



Mini Toolbar: Close to the cursor

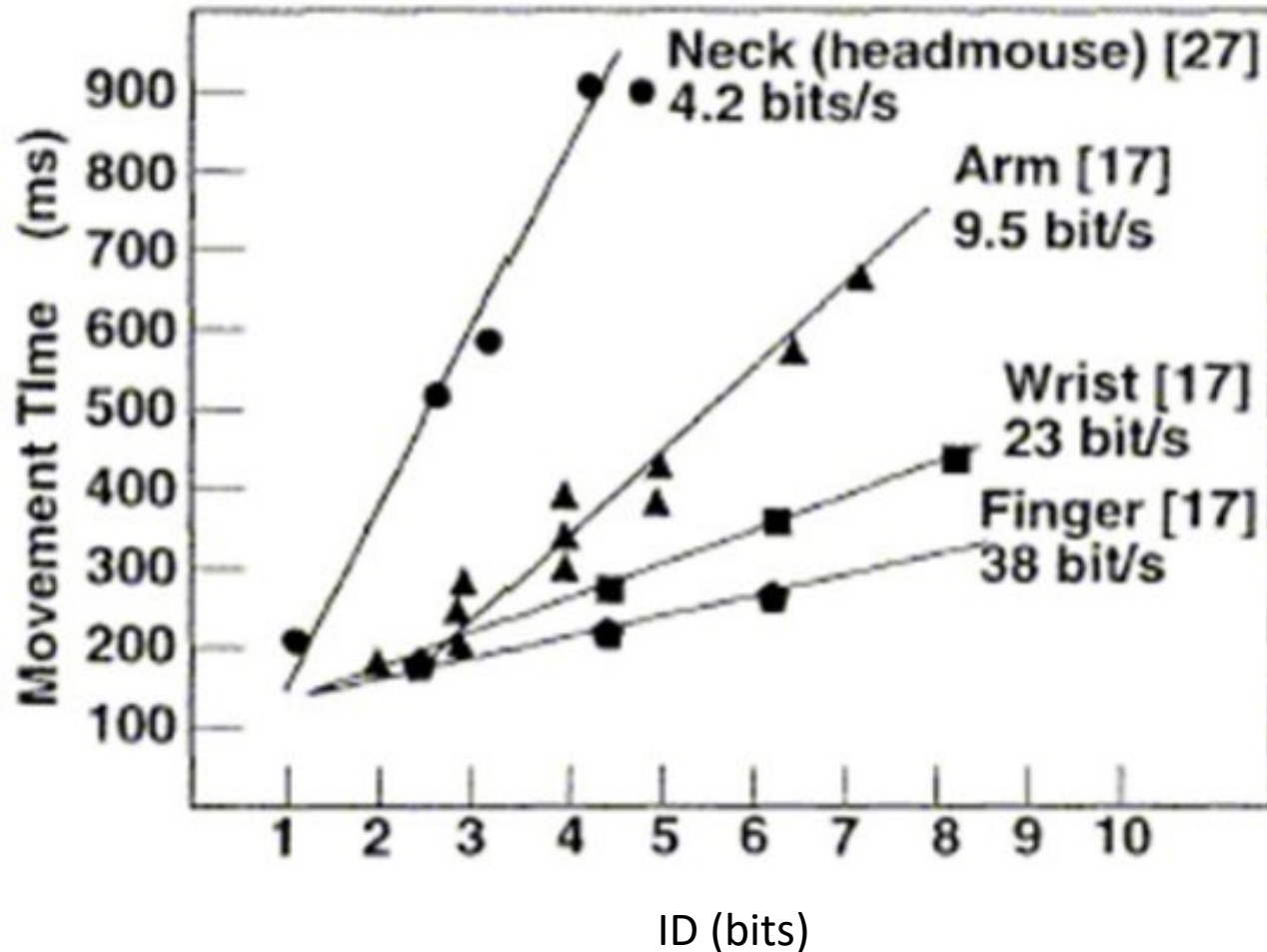


Magic Corner: Office Button in the upper-left corner

Source: Jensen Harris, An Office User Interface Blog : Giving You Fitts. Microsoft, 2007.

Bandwidth (IP) of Human Muscle Groups

$$MT = a + b \cdot ID \quad IP = 1/b$$



Which is faster?



50 years of data

Device	Study	<i>IP</i> (bits/s)
Hand	Fitts (1954)	10.6
Mouse	Card, English, & Burr (1978)	10.4
Joystick	Card, English, & Burr (1978)	5.0
Trackball	Epps (1986)	2.9
Touchpad	Epps (1986)	1.6
Eyetracker	Ware & Mikaelian (1987)	13.7

Reference:

Mackenzie, I. Fitts' Law as a research and design tool in human computer interaction. *Human Computer Interaction*, 1992, Vol. 7, pp. 91-139

Extending Fitts' law to Two-Dimensional Tasks

- Original Fitts' Law

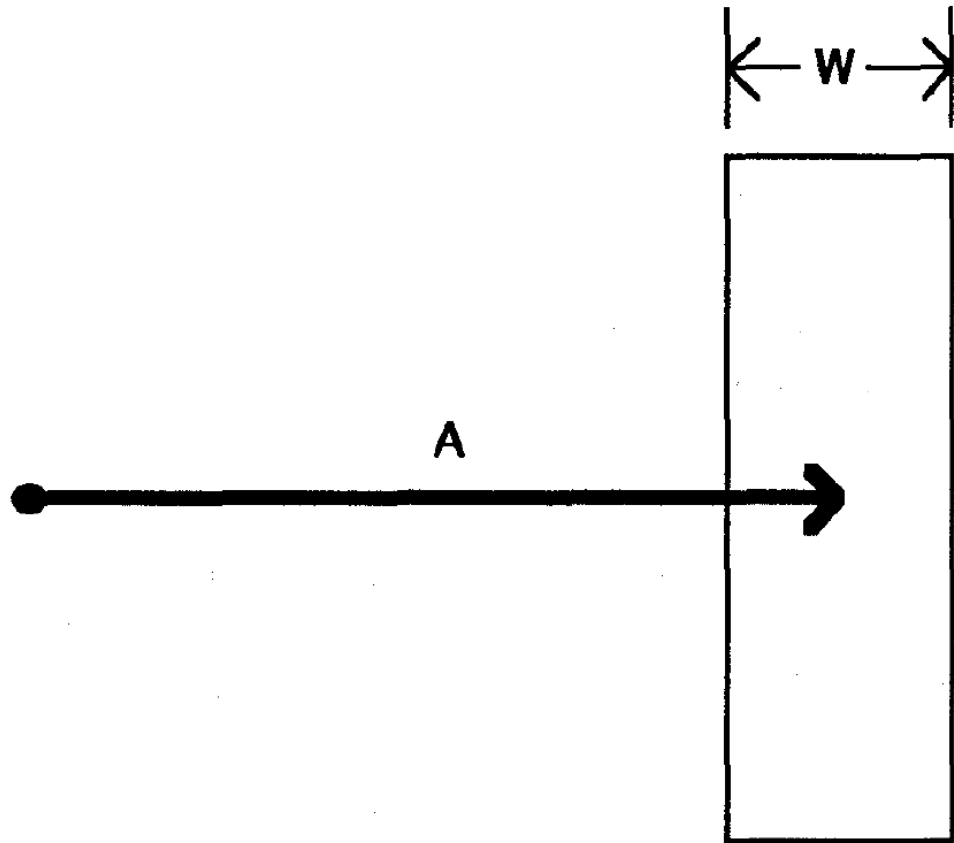
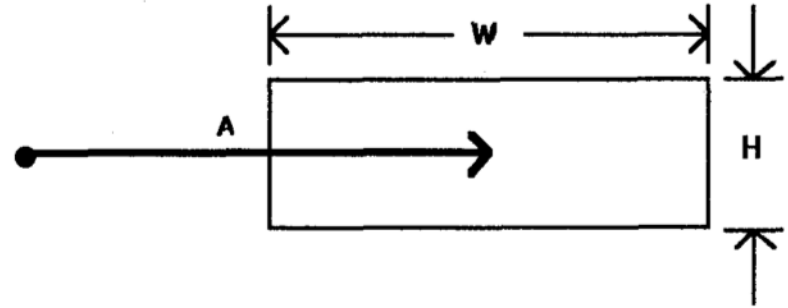


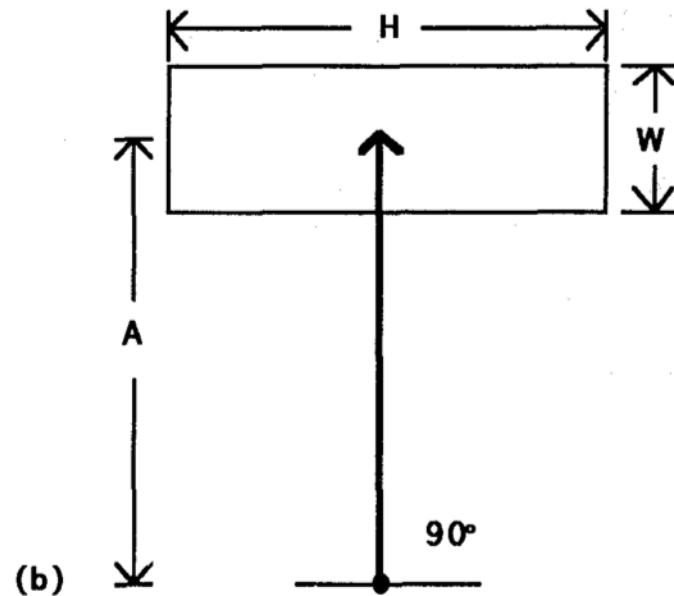
Figure 1. Fitts' law paradigm. The law is inherently one-dimensional since target amplitude (A) and width (W) are measured along the same axis.

Extending Fitts' law to Two-Dimensional Tasks

- Which one should we take into account? Width? Or Height?



(a)



(b)

Extending Fitts' law to Two-Dimensional Tasks

- How about the cases which are not horizontal or vertical?

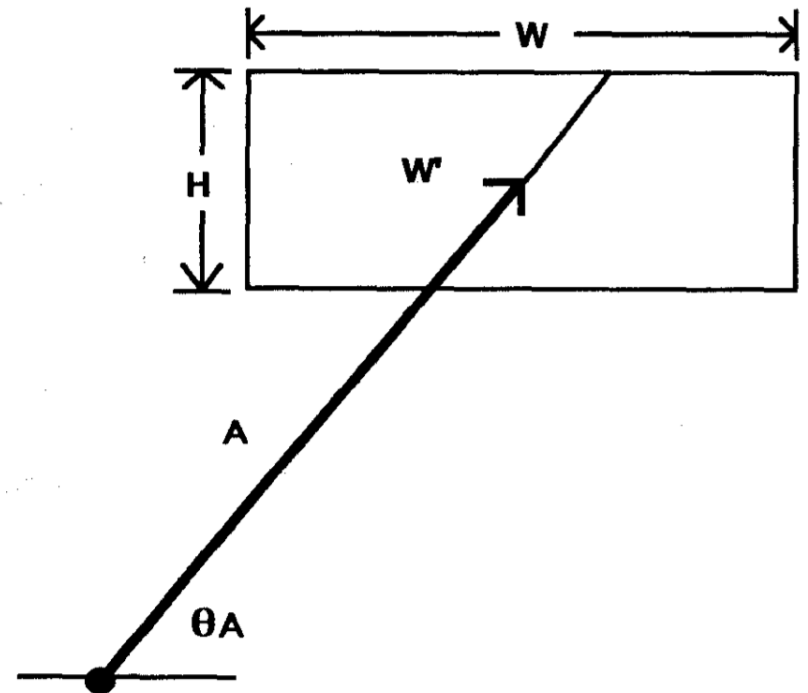


Figure 3. What is target width? Possibilities include W' (the width of the target along an approach vector) or the smaller of W or H .

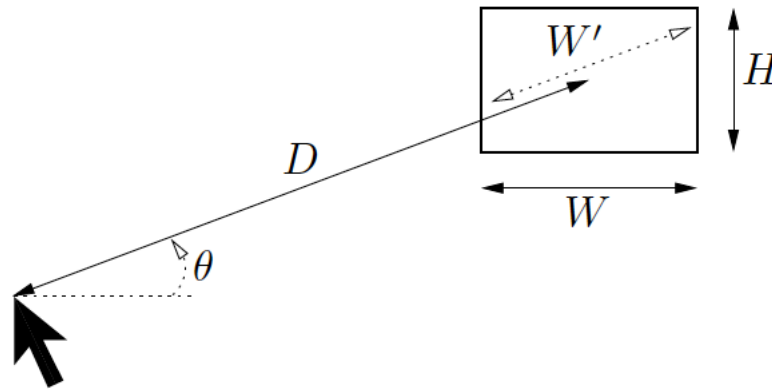
Extending Fitts' law to Two-Dimensional Tasks

- 5 model proposed
 - “STATUS QUO” model
 - “SMALLER OF” model
 - Apparent width W' model
 - Substitute W with $W*H$
 - Substitute W with $W+H$

$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

$$ID_{\min(W,H)} = \log_2 \left(\frac{D}{\min(W,H)} + 1 \right)$$

$$ID_{W'} = \log_2 \left(\frac{D}{W'} + 1 \right)$$



Extending Fitts' law to Two-Dimensional Tasks

- Experiment Result

Empirically, this is the best

Model for Target Width	<u>ID Range (bits)</u>		r^a	SE^b (ms)	<u>Regression Coefficients</u>		
	Low	High			Intercept, a (ms)	Slope, b (ms/bit)	IP (bits/s)
SMALLER-OF	1.58	5.04	.9501	64	230	166	6.0
W^1	1.00	5.04	.9333	74	337	160	6.3
$W+H$	0.74	3.54	.8755	99	402	218	4.6
$W \times H$	0.32	4.09	.8446	110	481	173	5.8
STATUS QUO	1.00	5.04	.8097	121	409	135	7.4

^a $n = 78, p < .001$

^bstandard error of estimate

Theoretically, this is similar to Fitts' Law

Figure 6. Correlations and regression coefficients for five models for target width.

Today's Questions

You have a palette of tools in a graphics application that consists of a matrix of 20x16-pixel icons laid out as a 2x8 array that lies along the left-hand edge of the screen. Without moving the array from the left-hand side of the screen or changing the size of the icons, what steps can you take to decrease the time necessary to access the average tool?



Refining Fitts' law models for bivariate pointing

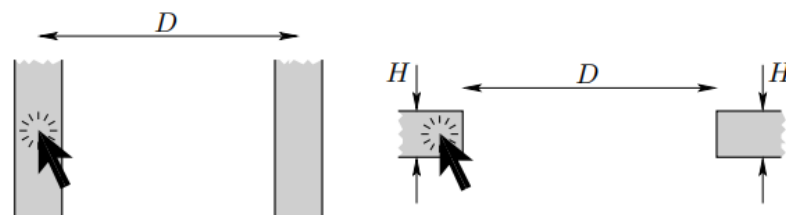
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ABSTRACT

We investigate bivariate pointing in light of the recent progress in the modeling of univariate pointing. Unlike previous studies, we focus on the effect of target shape (width and height ratio) on pointing performance, particularly when such a ratio is between 1 and 2. Results showed unequal impact of amplitude and directional constraints, with the former dominating the latter. Investigating models based on the notion of weighted ℓ_p norm, we found that our empirical findings were best captured by an Euclidean model with one free weight. This model significantly outperforms the best model to date.

in Fitts' law studies [7], and the usual practice is to take the diameter of the inscribed circle as the W in Fitts' law. In fact, two of the three tasks (peg in hole and disk transfer) in Fitts' original study [4] involved circle-shaped targets.



Define the appropriate “distance” in a 2D space.

$$X = \left(\frac{D}{W}, \frac{D}{H} \right)$$

$$\|X\|_1 = \frac{D}{W} + \frac{D}{H}$$

$$\|X\|_2 = \sqrt{\left(\frac{D}{W} \right)^2 + \left(\frac{D}{H} \right)^2}$$

the general weighted ℓ_p -norm model

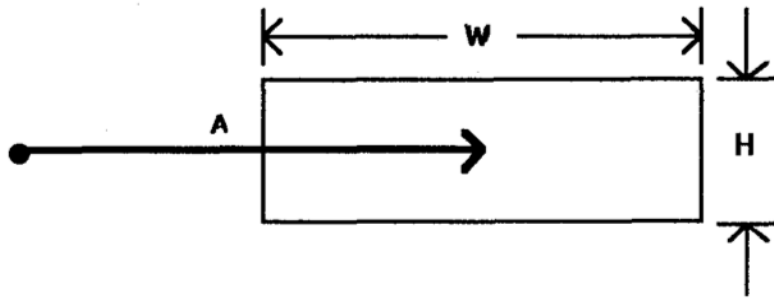
$$T = a + b \log_2 \left(\left[\omega \left(\frac{D}{W} \right)^p + \eta \left(\frac{D}{H} \right)^p \right]^{1/p} + 1 \right)$$

The best one:

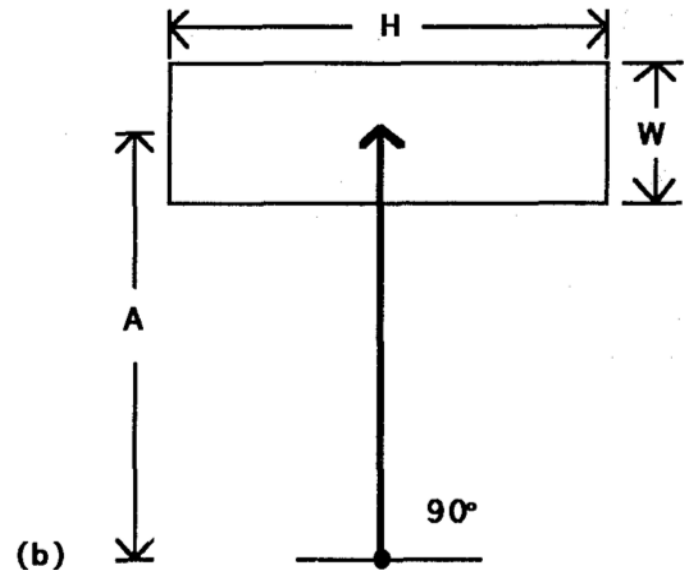
$$T = a + b \log_2 \left(\sqrt{\left(\frac{D}{W} \right)^2 + \eta \left(\frac{D}{H} \right)^2} + 1 \right)$$

$$T = a + b \log_2 \left(\sqrt{\left(\frac{D}{W}\right)^2 + \eta \left(\frac{D}{H}\right)^2} + 1 \right)$$

η in $[1/7, 1/3]$

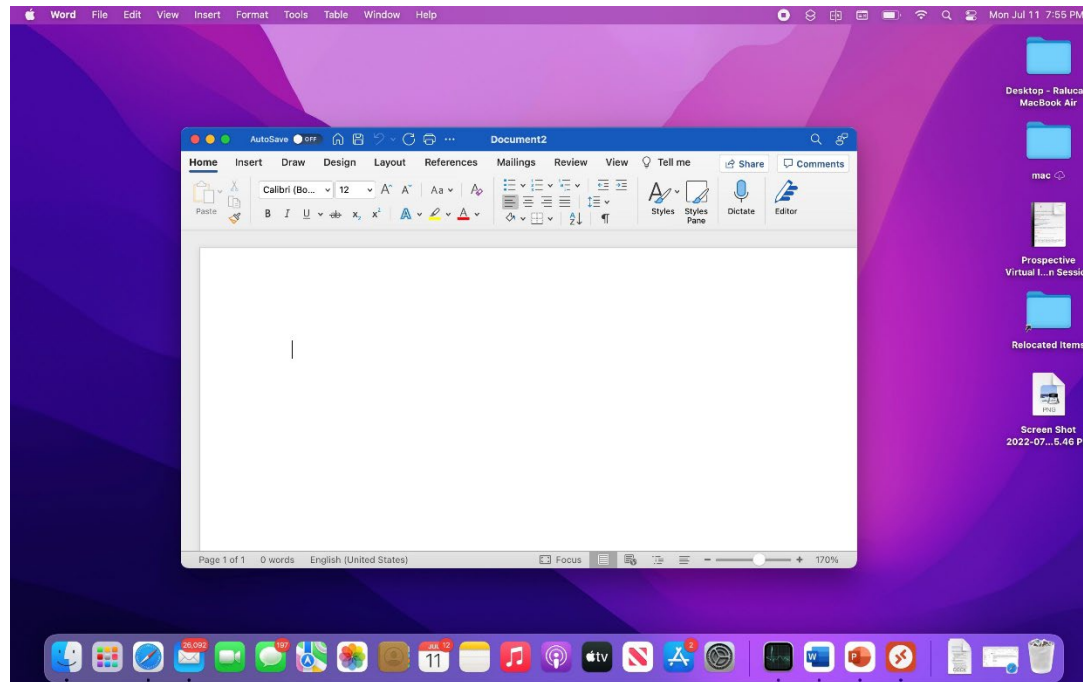


(a)



(b)

Application of Fitts' Law in UI Design



The application menu is positioned at the top edge of the screen, creating an infinite edge that allows faster access.

Application of Fitts' Law in UI Design

Icons Plus Labels

